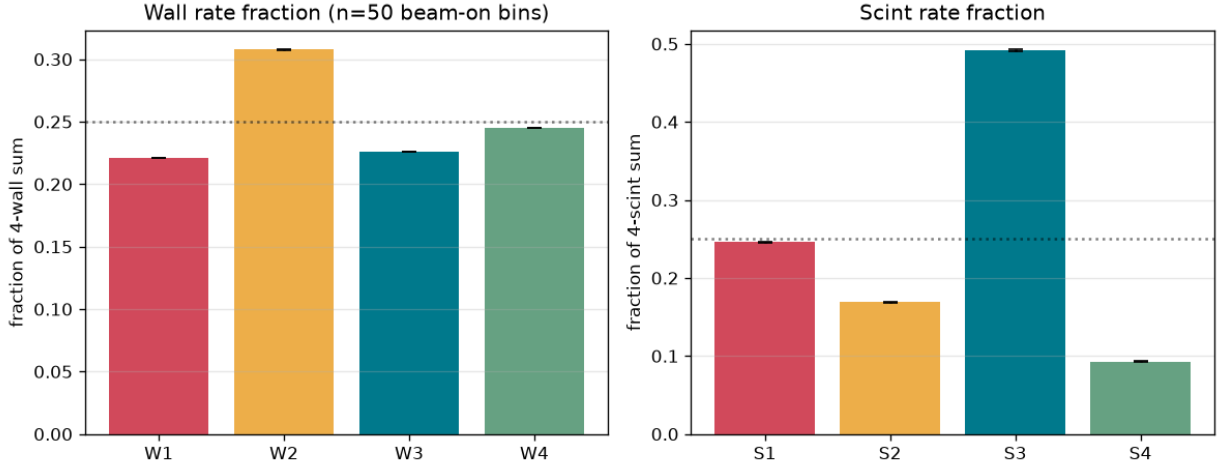


n_TOF X17 — N1081B Trigger Rate Measurements

2026-07-11 (FEU DAQ offline; parasitic beam)

1. Element-to-element rate variation (beam-normalized)

Phase 1: 50 beam-on 30 s bins. Each channel's rate as a fraction of its 4-channel sum (beam-independent). Uniform would be 0.25 each (dotted line).

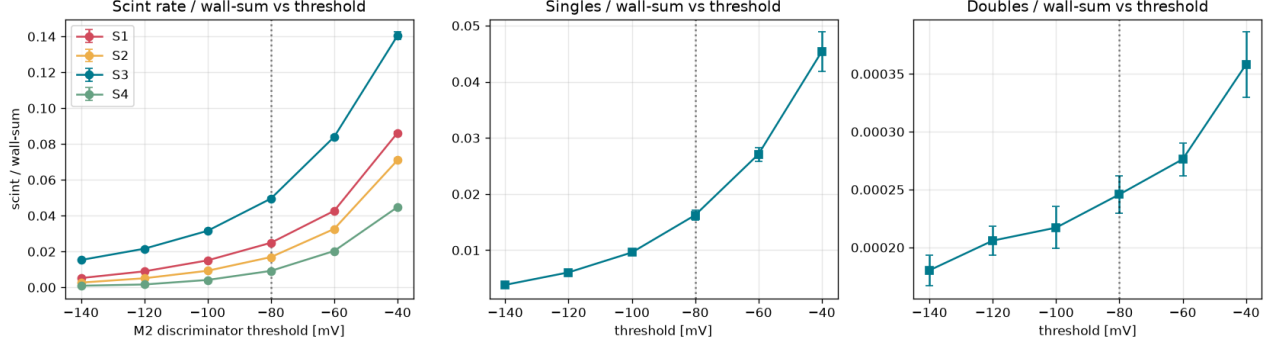


	W1	W2	W3	W4
Wall fraction	0.22	0.31	0.23	0.25
Wall rate (Hz)	652	907	667	723
	S1	S2	S3	S4
Scint fraction	0.25	0.17	0.49	0.09
Scint rate (Hz)	73	51	147	28

Hottest wall: **W2** ($1.39\times$ the coldest). Hottest scint: **S3** ($5.30\times$ the coldest).

2. Scint threshold scan (walls as beam monitor)

Phase 2: all four M2 scint sections set to a common discriminator threshold; rates normalized by the wall sum (threshold-independent). Nominal -80 mV marked (dotted).



Recommended equalizing thresholds (bring each scint to the *median* of the four wall-normalized rates at -80 mV; hot channels move to higher $|\text{threshold}|$, cold to lower; *not applied* — config left at -80 mV):

	S1	S2	S3	S4
Rec. threshold (mV)	-88.12	-74.93	-122.17	-59.53

3. Singles & Doubles trigger rates (good statistics)

Phase 3: 50 beam-on 60 s bins, nominal thresholds, beam $\sim 708 \times 10^{10}$ protons.

Signal	Rate (Hz)	Poisson err	counts
Singles (M4.A)	49.40	± 0.13	148268
Doubles (M4.B)	0.712	± 0.015	2136

Doubles/Singles = 0.0144. Integration time 50 min (beam-on).

4. Beam-off baseline & beam-induced rates

8 beam-off 30 s bins vs. the beam-on means above. "Beam-induced" = on – off.

	Beam-off (Hz)	Beam-on (Hz)	Beam-induced (Hz)
Wall W1	66	652	586
Wall W2	92	907	814
Wall W3	65	667	602
Wall W4	79	723	643
Scint S1	7.6	73	66
Scint S2	6.4	51	44
Scint S3	8.7	147	138
Scint S4	5.0	28	23
Singles	11.8	49.4	37.6
Doubles	0.112	0.712	0.599

Interpretation:

- **W2's excess is intrinsic, not beam.** $W2/W1 = 1.41 \times$ beam-off and $1.39 \times$ beam-on — the same multiplicative offset, so it is a detector property (SiPM dark/cosmic rate), not beam geometry.

- **S3's excess is beam flux, not noise.** Beam-off the scints are nearly uniform ($S3/S4 = 1.7\times$), but the beam-induced part makes $S3/S4 = 6.1\times$ — S3 sees genuinely more beam. Equalizing it at the discriminator would cut *real signal*, not noise.
- Cosmic/accidental floor: 16% of the beam-on Doubles (0.112 of 0.712 Hz) is present with no beam.

Data: `rate_campaign_run1.json`; plots `rate_ratios.png`, `threshold_scan.png`. Walls measured via M5.A (M1 outputs still flow though M1 is network-offline); scint thresholds set per M2 section.